

Benchmark Contamination as a Privacy/Security Vulnerability in Large Language Models (working draft, Pythia-160m)

Large language models are ranked and certified as safe on public benchmarks whose validity rests on the benchmark not appearing in pre-training. We study *benchmark contamination* not as a measurement-hygiene problem but as a privacy/security vulnerability: contamination is a visible symptom of memorization, and memorization is the mechanism by which sensitive content leaks. Using the Pythia suite trained on the public Pile—so that membership is ground truth rather than an inferred label—we run a systematic, pre-registered comparison of existing contamination/membership detectors (LOSS, Min-K%, Min-K%++, zlib) against a per-item *extraction* outcome. We make no claim to a new detector or metric. Our contribution is a *controlled* result: a pre-registered partial-correlation and mediation analysis that isolates the role of raw per-item loss. We find that the apparent contamination→leakage association is *loss-mediated to the resolution of this experiment*: the calibrated reference-free detectors—which are themselves strongly-to-moderately collinear with loss (Spearman 0.74–0.90)—add no positive predictive value beyond it, and their residual partials are null or, for the most loss-collinear detector (Min-K%), weakly negative in a manner consistent with a suppression artifact rather than substantive inverse prediction. This absence of positive residual survives a non-linear (cubic-residual and decile-stratified) loss control and deduplication, and is not explained by token frequency or the zero-inflated outcome. We frame this as a *membership-detection-versus-leakage-prediction divergence*: the detectors the field optimizes for membership are not the right instrument for the privacy question. **These results are preliminary, obtained on the smallest (160M) Pythia model on CPU; the pipeline is built so that the GPU-scale replication is a single configuration change.** All analyses are pre-registered and every number is reproducible from a seeded script.

Introduction

Large language models (LLMs) are ranked, selected, and certified as safe largely on the basis of their scores on public benchmarks [hendrycks2021mmlu,cobbe2021gsm8k]. Those scores are only meaningful under one assumption: that the evaluation data was absent from pre-training. The assumption is increasingly untenable. Benchmarks are small, static, and endlessly redistributed across the web, while training corpora are weakly filtered crawls assembled at the scale of hundreds of gigabytes to petabytes [commoncrawl,gao2020pile]; benchmark items are therefore swept into the next crawl by ordinary copying, with no adversary required. The resulting *benchmark contamination*—the presence of evaluation data in the training corpus [golchin2024timetravel]—is usually treated as a measurement-hygiene problem: a contaminated score over-states capability [ravaut2024survey].

We argue that contamination is better understood as a *privacy and security* vulnerability, and we study it as one. The same over-parameterized models that score highly on a leaked benchmark also memorize and can regurgitate verbatim training sequences, including personally identifiable information (PII) that co-occurs in the same corpora [carlini2021extracting,carlini2023quantifying]. Contamination, in this view, is a visible symptom of unintended memorization, and memorization is the mechanism by which sensitive content leaks. If a cheap, model-side contamination signal predicts which items the model has memorized, then the act of contaminating a benchmark is not merely inflating a metric; it is

exposing a leakage channel. We make this contamination → memorization → leakage chain the object of empirical study, on models whose training corpus is fully public so that membership is ground truth rather than a guess.

The privacy/security community has, however, established that membership signal on pre-trained LLMs is weak: large-scale audits on the Pythia suite and The Pile report that membership-inference attacks (MIAs) barely exceed chance, and that apparent successes often reflect distribution shift between the member and non-member sets rather than membership itself [duan2024mia]. We take this finding as a constraint, not an obstacle. Rather than claim a stronger attack, we ask a sharper, security-relevant question: *even where the membership signal is weak, does it still predict concrete leakage?* Answering it requires the evaluation discipline that security venues expect of a privacy attack—true-positive rate at a low, fixed false-positive rate, read off a log-scale ROC curve, rather than an average-case AUC that hides whether the attack ever fires confidently [carlini2022lira]. It also exposes a question the membership-inference literature does not ask: detectors are tuned and ranked by how well they separate members from non-members, but leakage is a property of *how much* the model memorized a specific item. We therefore evaluate each detector not only as a membership classifier but as a predictor of concrete leakage, and ask whether the two objectives coincide—finding that they do not.

Contributions (and explicit non-contributions).

We are deliberate about what this paper is and is not. It is *not* a new detector, attack, or metric: every detection method we run is from prior work [yeom2018privacy,shi2024detecting,zhang2025minkpp,carlini2021extracting,brown2020gpt3,oren2024proving], and our evaluation protocol is the established low-FPR convention of Carlini et al. [carlini2022lira]. Within that honest scope, our contributions are:

- **A security reframing and threat model.** We recast benchmark contamination as a membership/exposure vulnerability with an explicit adversary and graded goals—membership inference on a single item, benchmark-level contamination confirmation, and verbatim/PII extraction—rather than as a measurement artifact (Section [2](#sec:background), Section [5](#sec:eval)).
- **A systematic comparative evaluation of existing detectors under the S&P low-FPR protocol on ground-truth Pile membership.** We evaluate LOSS/perplexity, Min-K%, Min-K%++, and the zlib ratio as membership detectors, the corpus-side n -gram overlap test as a contamination-label oracle, and the Oren permutation/exchangeability test at the benchmark level, all on Pythia trained on the public Pile, reporting TPR at 0.1% and 1% FPR with log-scale ROC and bootstrap confidence intervals, with explicit controls for the frequency, duplication, and temporal confounds that prior work identifies [biderman2023pythia,gao2020pile,duan2024mia].
- **A pre-registered measurement of *which* contamination signal predicts leakage—and which does not.** We correlate per-item contamination scores against an extraction outcome—prefix-continuation extractable memorization under greedy decoding [carlini2023quantifying]—and, on the Enron Emails subset that already sits inside the Pile, against regex-detected PII leakage [lukas2023pii]. A pre-registered partial-correlation and mediation control then isolates the role of raw loss. In our ground-truth 160M-parameter setting we find that the contamination→leakage association

is *loss-mediated to the resolution of this experiment*: once loss is held fixed, the calibrated reference-free detectors (Min-K%, Min-K%+, zlib) add no positive predictive value. These detectors are themselves near-collinear transforms of loss (Spearman 0.74–0.90), so we read a negative residual for the most collinear of them (Min-K%) as a likely suppression artifact rather than substantive inverse prediction, and claim only the conservative result: no positive signal beyond loss. The calibrations that improve membership-detection AUC thus do not retain the loss-magnitude signal that predicts leakage—a divergence between membership detection and leakage prediction that we report as our central empirical finding (robust to deduplication and a non-linear loss control, and not explained by token frequency or the zero-inflated outcome).

We do not propose internal-probe or other novel detectors as contributions, do not train or fine-tune models, and do not attack closed production systems for real third-party PII; differential privacy and related defenses are discussed as the mitigation direction only.

Background: LLM Evaluation Benchmarks as an Attack Surface

Benchmarks as proxies for latent capabilities

Large language model (LLM) benchmarks function as *proxies* for latent capabilities—reasoning, comprehension, factual knowledge, coding proficiency—that cannot be measured directly. By scoring a model on a fixed set of standardized tasks, the community infers a model’s likely utility (and, increasingly, its safety) in deployment [hendrycks2021mmlu]. Canonical examples target distinct competencies: MMLU for broad multitask knowledge across 57 subjects [hendrycks2021mmlu], GSM8K for multi-step mathematical reasoning [cobbe2021gsm8k], and HumanEval for functional code generation [chen2021humaneval]. Reported scores on these suites drive model-selection decisions, leaderboard rankings, and published claims of progress.

The core validity assumption

The inferential validity of benchmark evaluation rests on one strict assumption: *the test data was not seen during pre-training*. Only under this assumption does high benchmark performance license the intended conclusion—that the model *generalizes* (applies learned regularities to novel inputs) rather than *memorizes* (retrieves specific training instances). When the assumption is violated, the benchmark no longer measures capability; a memorized test item inflates the score without any corresponding gain in generalization, rendering the metric an unreliable estimator of the construct it claims to measure. The generalization-versus-memorization distinction is not merely conceptual: memorization is directly measurable as the verbatim regeneration of training sequences and grows predictably—log-linearly in model scale, data duplication, and context length [carlini2023quantifying]. The same phenomenon has a sharper, privacy-relevant form: a planted secret’s *exposure*, the model’s tendency to rank that secret above random alternatives, rises with how often it was seen during training [carlini2019secret], and which specific examples a model memorizes is itself a measurable, example-level property rather than a uniform background rate [zhang2023counterfactual]. A memorized benchmark item is thus the visible end of the same mechanism that retains rare, sensitive strings.

Static test sets meet weakly filtered corpora

The security-relevant tension is structural. Evaluation benchmarks are *static, small, widely circulated, and publicly indexed*: once published, an MMLU or GSM8K item is copied into

papers, blog posts, GitHub repositories, and discussion forums. Training corpora, by contrast, are *massive web scrapes with weak filtering*—Common Crawl [commoncrawl] and The Pile [gao2020pile] are assembled at the scale of hundreds of gigabytes to petabytes, where exhaustive removal of any particular short string is impractical. The natural consequence is that benchmark items are swept into training corpora through ordinary web redistribution, with no adversary required. This makes a public benchmark a persistent, low-effort *attack surface*: the same property that makes a benchmark useful (stable, shared, citable) is what guarantees its eventual presence in the next corpus crawl.

We argue this is best understood through a security lens rather than purely as a measurement-hygiene problem. Contamination converts an evaluation artifact into a channel that (i) invalidates the safety and capability claims downstream decisions rely on, and (ii)—the focus of this paper—couples directly to *memorization*, and through memorization to the leakage of sensitive content that co-occurs in the same weakly filtered corpora. Section [4](#sec:relatedwork) formalizes contamination, its typology, and the detection and memorization literature on which our evaluation builds.

Threat Model

We frame contamination detection as a membership/exposure attack and state the adversary explicitly, following the convention that a privacy attack must be evaluated by its behaviour at a low false-positive operating point rather than on average [carlini2022lira].

Adversary goals (graded).

- **G1 — Membership inference.** Decide whether a specific sequence (a benchmark item, document, or record) was in the training corpus.
- **G2 — Benchmark-level contamination confirmation.** Decide, with a controlled false-positive rate, whether an entire benchmark was trained on.
- **G3 — Extraction / leakage.** Recover verbatim content—and, on a controlled corpus, PII—that was in training. This is the concrete harm; G1-G2 are of interest largely insofar as they predict G3.

Adversary knowledge and access.

We grade detectors by the minimum access each requires: *black-box* (text in, text out; e.g. guided prompting, which we do not evaluate), *gray-box* (per-token log-probabilities / loss; LOSS, Min-K%, zlib), and *white-box* (the full next-token distribution; Min-K%+). The corpus-side *n*-gram test instead assumes access to the training corpus (available here because the Pile is public) and is used to construct ground-truth contamination labels, not as a model-access attack. For our ground-truth experiments the *auditor* additionally knows the public training corpus; the modelled attacker does not need corpus access for G1/G3.

Success criteria.

G1: true-positive rate at 0.1% and 1% false-positive rate (log-scale ROC), with AUC secondary and bootstrap confidence intervals. G2: a permutation-test *p*-value below threshold with a controlled false-positive rate [oren2024proving]. G3: a non-zero extraction rate, and—our headline analysis—a positive association between a per-item contamination score and the per-item extraction outcome that *survives controlling for raw loss*. The last criterion is what distinguishes a contamination signal that genuinely predicts leakage from one that merely restates the model’s loss.

Out of scope.

We do not attack closed production models for real third-party PII, do not train or fine-tune models, and propose no new detector. Differential privacy is discussed as the producer-side mitigation our threat model motivates (Section [4.5](#sec:dp)), not implemented.

Related Work: Contamination, Memorization, and Privacy Leakage

Defining benchmark contamination

We adopt the standard definition: *benchmark contamination* is the presence of evaluation data—inputs, labels, or accompanying metadata—within a model’s pre-training corpus [golchin2024timetravel]. Contamination matters for two reasons that this paper treats as inseparable. First, it invalidates evaluation: a contaminated score conflates capability with retrieval, so the metric no longer estimates generalization. Second, and central to our thesis, contamination is a *symptom of, and a measurable proxy for, unintended memorization*—and memorization of evaluation data sits on the same mechanism that leaks sensitive content from the corpus. We make this contamination → memorization → leakage chain the object of empirical study.

A typology of contamination

Following the project’s framing and the contamination-detection survey [ravaut2024survey], we distinguish three forms by the transformation between the corpus copy and the benchmark item:

- **Verbatim contamination.** The exact token sequence of a test item appears in training data. This is what classical n -gram decontamination targets (e.g., the 13-gram overlap test introduced for GPT-3 [brown2020gpt3]) and what verbatim-extraction memorization measures [carlini2023quantifying].
- **Paraphrased contamination.** The semantic content is present but reworded, so surface-level n -gram matching misses it. A perfect verbatim filter provides only a false sense of safety, since style-transfer rephrasings evade it while preserving the leaked information [ippolito2023verbatim].
- **Semantic contamination.** The underlying knowledge or answer is encoded without lexical overlap (e.g., the same question-answer mapping in a different format). Detecting it requires model-behavioral or distributional signals rather than string matching.

A second, orthogonal severity axis is *what* is contaminated: input-only leakage inflates familiarity, whereas joint input-label leakage enables direct answer retrieval and is the most damaging to evaluation validity. Empirically, overlap between open-model training data and benchmarks such as GSM8K has been reported for models trained on largely undisclosed corpora [touvron2023llama], motivating ground-truth-controlled study on models whose corpus is fully public.

Why memorization is a security and privacy problem

Memorization is not a benign curiosity. Over-parameterized models trained on web-scale scrapes retain and can regurgitate verbatim sequences, including personally identifiable information (PII) such as names, emails, and phone numbers [carlini2021extracting]. This has

been formalized along several axes that we reuse as outcome variables:

- ***k*-eidetic / extractable memorization.** A string is extractable if a prefix makes the model regenerate it, and is *k*-eidetic if it occurs in at most *k* training documents [carlini2021extracting]; the prefix-continuation form under greedy decoding makes this directly measurable [carlini2023quantifying].
- **Exposure and example-level memorization.** Injecting a canary secret and measuring its *exposure*—its rank against random alternatives—quantifies unintended memorization and its growth with occurrence count [carlini2019secret]; this requires control over the training process (canary insertion), which our pretrained-checkpoint setting does not afford, so we use it for definitions rather than as a measurement. Relatedly, memorization is concentrated on specific examples [zhang2023counterfactual] rather than spread uniformly, which is what makes per-item contamination scores meaningful predictors of per-item leakage.
- **Extraction at scale.** Production models can be driven, via a divergence attack, to emit memorized training data well above their nominal aligned rate, recovering thousands of verbatim examples cheaply [nasr2025scalable].
- **PII leakage games.** Leakage of personally identifiable information decomposes into extraction, reconstruction, and inference; data scrubbing and differential privacy reduce but do not eliminate it [lukas2023pii], models leak PII through memorization more than through associative inference [huang2022leaking], and black-box probing tools can elicit a data subject’s PII directly from a deployed model [kim2023propile].

The security framing follows directly: if contamination is a measurable proxy for memorization, and memorization is the vector for PII and proprietary-data exposure, then contamination is not only a metrics problem but a *privacy vulnerability*.

The membership-inference lineage

Deciding whether a specific record was in a model’s training set is the canonical privacy attack, and contamination detection is an instance of it. The lineage we build on runs as follows. *Shadow-model* attacks established the threat: by training reference models on data drawn from the same distribution, an adversary learns to distinguish members from non-members from the target model’s outputs [shokri2017membership]. Yeom et al. tied attack success to overfitting and gave the simplest practical baseline—threshold the per-example loss—together with the *membership advantage* (TPR–FPR) figure of merit [yeom2018privacy]. Carlini et al.’s *Likelihood Ratio Attack* (LiRA) then reframed MIA from first principles as a per-example hypothesis test calibrated with shadow models, and—central to our methodology—argued that average-case AUC is the wrong yardstick for a privacy threat: an attack matters if it identifies *some* members with very few false accusations, so the right report is TPR at a low, fixed FPR on a log-scale ROC curve [carlini2022lira]. Shadow-model calibration, however, is infeasible at Pile/Pythia scale (it requires training many models on the training distribution), so we adopt LiRA’s *metric* but not its *attack*.

For pre-trained LLMs, the field moved to *reference-free* likelihood signals that need no shadow models. Min-K% Prob averages the log-probabilities of a sequence’s lowest-probability *k*% of tokens, on the hypothesis that members lack high-surprise outlier tokens [shi2024detecting]; Min-K%++ sharpens this by z-scoring each token against the *full* next-token distribution before averaging, detecting that the target token sits at a local maximum of the modeled distribution [zhang2025minkpp]. A parallel reference-free line, neighbourhood comparison,

calibrates a sample’s score against synthetically generated neighbour texts instead of a reference model [mattern2023neighbourhood]; we treat it as a related approach we do not evaluate, since it needs many extra masked-LM forward passes per example and, in the regime below, underperforms. The reality check on this whole line is the MIMIR study: a large-scale audit on Pythia (160M–12B) and The Pile with controlled member/non-member splits finds that these attacks barely exceed chance ($AUC \approx 0.5\text{--}0.6$), that LLMs see their corpus for too few epochs over too large a dataset to memorize in the way classical MIA assumes, and that apparent successes frequently reflect a temporal or topical *distribution shift* between the splits rather than membership [duan2024mia]. This finding defines our honesty constraint: we do not claim to beat these numbers; we ask whether the weak signal that remains still predicts leakage.

Differential privacy as the defense direction

The standard principled mitigation for training-data leakage is differential privacy. DP-SGD bounds any single example’s influence on the trained model by clipping per-example gradients and adding calibrated noise, with privacy accounted via the moments accountant [abadi2016deep]. Applied to language models, DP fine-tuning can retain much of the utility of non-private training, particularly with large pre-trained backbones [li2022dp.llm] and parameter-efficient adaptation [yu2022dp.finetuning]. DP bounds memorization and thereby the leakage we measure, but at a privacy-utility cost and—crucially for us—it must be applied *at training time*; it is a defense for model producers, not a detector available to an auditor of an already-released model. We therefore position DP as the mitigation our threat model motivates, and do not implement it (we train no models).

Existing detection techniques

We describe the techniques we implement and compare; the comparative evaluation and the access requirements appear in Section [5](#sec:eval). All operate without any novel detector of our own—our contribution is their security-framed, ground-truth evaluation, not a new method.

- ***n*-gram / substring overlap.** Flag a benchmark item that shares an *N*-gram with the corpus [brown2020gpt3]. Requires corpus access; misses paraphrased and semantic contamination.
- **Loss / perplexity thresholding.** The mandatory membership-inference baseline: members exhibit lower loss, with attack success tied to overfitting [yeom2018privacy].
- **Min-K% Prob.** Average the log-probabilities of the lowest-probability *k*% of tokens; reference-free and logprob-only [shi2024detecting].
- **Min-K%++.** Normalizes each token’s log-probability against the full next-token distribution before the bottom-*k*% average, the current state of the art among reference-free detectors [zhang2025minkpp].
- **zlib-entropy ratio.** Calibrate model perplexity by the zlib-compressed size of the text, controlling for intrinsic compressibility/frequency [carlini2021extracting].
- **Permutation / exchangeability test.** At the *benchmark* level rather than per item, score each ordering of a benchmark’s examples by the log-likelihood of their concatenation and compare the canonical (published) order against random shufflings; a model trained on the benchmark in canonical order favours it beyond chance, yielding a provable, FPR-controlled contamination certificate [oren2024proving].

We additionally note two techniques we describe but *do not* evaluate, since our ground-truth, logit-access setting makes likelihood-based detectors stronger and cleaner: *guided prompting*, which prompts a model with dataset metadata and a partial instance and tests for verbatim completion [golchin2024timetravel], a black-box signal aimed at closed models; and the reference-free *neighbourhood* and shadow-model *reference* attacks discussed in Section 4.4 [mattern2023neighbourhood,shokri2017membership].

Limitations of existing detection, and our positioning

Two limitations frame our contribution. First, *detection is fragile to the transformation*: string-matching misses paraphrased and semantic contamination [ippolito2023verbatim], and likelihood-based membership inference is known to barely exceed chance on pre-trained LLMs evaluated under controlled ground truth, because the corpora are seen for few epochs and member/non-member boundaries are fuzzy [duan2024mia]. Second, *evaluation conventions matter*: average-case AUC or accuracy can mask whether an attack confidently identifies any members, so the security-appropriate report is true-positive rate at low false-positive rate with log-scale ROC [carlini2022lira]. We therefore do not claim a stronger detector. We ask a different, security-relevant question: *even where contamination signal is weak, does it predict concrete privacy leakage?* We answer it with ground-truth membership on the Pythia suite [biderman2023pythia] trained on the public Pile [gao2020pile], under the low-FPR protocol, with explicit controls for the frequency, duplication, and temporal confounds that prior work identifies.

Closest prior work, and how we differ

Three recent works reach conclusions adjacent to ours, and we are careful to position against them rather than overclaim. Al Sahili et al. [alsahili2025effectiveness] reach a compatible conclusion for targeted extraction—that “complex MIA techniques yield only marginal improvements over simple likelihood-based ranking”—but they establish it through aggregate *ranking-precision* comparisons and an AdaBoost ensemble over MIA features, reporting *marginal gains* rather than testing for independent signal. In contrast, we run a pre-registered *partial correlation controlling for raw per-item loss*, which lets us state the stronger, calibrated claim that the reference-free detectors contribute *zero or negative* residual predictive value once loss is partialled out. Hayes et al. [hayes2025strong] likewise “observe no correlation with MIA success” for extraction and conclude the “two privacy attacks may capture different signals,” but their evidence is a *direct, zero-order* correlation between a reference-model attack (LiRA) and extraction. We differ on both method and object: we *partial out per-item loss* rather than correlating directly, and we target the reference-free *calibrated* detectors (Min-K%, Min-K%++, zlib) that the contamination-detection literature actually deploys, showing the divergence persists as a controlled mediation result. Independently, Chen et al. [chen2025statistical] find for the *membership* task that the few detectors numerically above the loss baseline (Min-K%, Min-K%++, ReCaLL) do not beat it robustly once random-seed variance is accounted for, and that performance is domain-dependent (code-like, low-token-diversity domains such as GitHub and StackExchange behave differently from Wikipedia and FreeLaw); we revisit this domain dependence for the *extraction* outcome in our per-domain analysis (Section 5), noting it is a distinct axis from their membership-AUC result. Finally, blind-baseline and SoK critiques [das2024blind,meeus2025sok] show that post-hoc member/non-member splits can make detector “success” an artifact of distribution shift; our use of ground-truth Pile membership (no post-hoc split) is precisely the design

discipline they call for.

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@p2.1cmp1.6cmp1.9cmp2.0cmp2.4cm@ **Study & Outcome & Detectors & Statistical method & Conclusion** Shi'24; Zhang'25 [shi2024detecting,zhang2025minkpp] & membership & reference-free (Min-K%/++) & AUC / TPR@FPR & detector raises membership AUC Duan'24 (MIMIR) [duan2024mia] & membership & ref-free + reference & AUC on ground truth & MIAs $\approx$ chance on LLMs Carlini'22 (LiRA) [carlini2022lira] & membership & shadow/reference & TPR at low FPR & strong only with shadow models Chen'25 [chen2025statistical] & membership & reference-free & seed-variance testing vs loss & not robustly beyond loss Hayes'25 [hayes2025strong] & membership & extraction & LiRA (reference) & direct (zero-order) correlation & MIA $\neq$ extraction Al Sahili'25 [alsahili2025effectiveness] & extraction (targeted) & ref-free + AdaBoost & ranking precision; ensemble & marginal gains over likelihood **This work & extraction & ref-free calibrated & partial corr. + mediation (control loss) & zero/negative residual beyond loss**

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Evaluation Overview

Threat model and success criteria

We frame contamination detection as a membership/exposure attack with an explicit adversary (Section omitted here; see docs/experiment_design.md). Goals range from membership inference on a single item, to benchmark-level contamination confirmation, to verbatim extraction and PII leakage. Each detector is evaluated at its minimum access tier (gray-box logprobs for LOSS/Min-K%/zlib; white-box logits for Min-K%++). Success is defined by the security-appropriate operating point rather than average accuracy.

Methods under comparison

We evaluate *existing* detectors only—we propose no new detector. The per-item membership suite is LOSS/perplexity [yeom2018privacy], Min-K% Prob [shi2024detecting], Min-K%++ [zhang2025minkpp], and the zlib-entropy ratio [carlini2021extracting]. Two further tests operate off the per-item likelihood axis: corpus-side n -gram overlap [brown2020gpt3], a model-free data-side check used to construct ground-truth contamination labels for benchmark items, and the Oren permutation/exchangeability test [oren2024proving], a benchmark-level test that compares the canonical ordering of a benchmark's examples against random shufflings to certify contamination with a controlled false-positive rate. The leakage outcome is prefix-continuation extractable memorization under greedy decoding [carlini2023quantifying]; on the controlled corpus we additionally measure regex-detected PII leakage, framed via the PII-leakage games of Lukas et al. [lukas2023pii]. Related approaches we deliberately *do not* evaluate—guided prompting [golchin2024timetravel], neighbourhood and shadow-model reference attacks [mattern2023neighbourhood,shokri2017membership], and the divergence-style extraction of production models [nasr2025scalable]—are discussed in Section [4. **[D1]** An internal-activation probe is reported, if at all, only as exploratory analysis in the Discussion, not as a contribution.

Data

Table [\[tab:datasets\]](#tab:datasets) summarizes every corpus and benchmark used or referenced below.

Models and corpus.

The primary model is the Pythia suite [biderman2023pythia], trained on the public Pile [gao2020pile]; its reconstructible training order, 154 checkpoints, multiple sizes, and deduplicated variant provide exact membership ground truth. We use the released MIMIR member/ non-member splits [duan2024mia], which control n -gram overlap between members and non-members. OLMO [groeneveld2024olmo] on Dolma [soldaini2024dolma] is a secondary replication target. The Pile sits within the broader weakly filtered web-scrape regime—Common Crawl [commoncrawl] and its filtered derivatives C4 [raffel2020c4,dodge2021c4] and RedPajama [weber2024redpajama]—that makes benchmark contamination structural rather than adversarial.

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Dataset	Type	What it is	Size	Cite
The Pile	corpus	Curated 22-subset English corpus; Pythia’s training data and our membership ground truth	825 GB	[gao2020pile]
Common Crawl	corpus	Open, continually updated repository of raw web-crawl data; the base of most LLM pre-training scrapes	petabyte-scale (growing)	[commoncrawl]
C4	corpus	Colossal Clean Crawled Corpus: a filtered Common Crawl snapshot introduced with T5	~750 GB	[raffel2020c4,dodge2021c4]
Dolma	corpus	Open pre-training corpus; OLMO’s training data (replication target)	3 T tokens	[soldaini2024dolma]
RedPajama	corpus	Open reproduction of an LLaMA-style pre-training mixture	~30 T tokens	[weber2024redpajama]
MMLU	benchmark	Multiple-choice knowledge/reasoning across 57 subjects	15,908 questions	[hendrycks2021mmlu]
GSM8K	benchmark	Grade-school multi-step math word problems	8,500 problems	[cobbe2021gsm8k]
HumanEval	benchmark	Hand-written Python programming problems with unit tests	164 problems	[chen2021humaneval]
HellaSwag	benchmark	Adversarially filtered commonsense sentence completion	~70,000 items	[zellers2019hellaswag]
TruthfulQA	benchmark	Questions probing imitative falsehoods	817 questions	[lin2022truthfulqa]
BoolQ	benchmark	Naturally occurring yes/no reading-comprehension questions	15,942 questions	[clark2019boolq]

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Benchmarks and PII.

Contamination is tested against MMLU, GSM8K, HumanEval, HellaSwag, TruthfulQA, and BoolQ.

[D3] For PII leakage we use the Enron Emails data *as a Pile subset already present in Pythia’s training data*, plus a synthetic PII set for controlled structure, rather than fine-tuning a model to memorize PII. All PII results are reported in aggregate; no real PII is reproduced in the paper.

Metrics (each justified)

[D2] Following the membership-inference-from-first-principles convention [carlini2022lira], the primary metric is *true-positive rate at a fixed low false-positive rate* (TPR @ 0.1% and 1% FPR) reported with *log-scale ROC*; AUC-ROC is reported secondarily. These capture whether a detector *confidently* identifies members—the privacy-relevant regime—which average-case accuracy hides. For benchmark flagging at a chosen operating threshold we additionally report precision/recall/F1 as a secondary, application-facing view. The leakage outcome is the *extraction rate* [carlini2023quantifying]. The headline analysis is the *Spearman correlation between per-item contamination score and per-item extraction/leakage outcome*, with bootstrap confidence intervals and a pre-registered partial-correlation control that isolates the contribution of raw loss—the quantitative form of the paper’s central question.

Validation and controls

[D4] Robustness is established by repeating each measurement over multiple seeds with bootstrap confidence intervals on TPR@FPR and on the Spearman correlation, and by a permutation/exchangeability test for benchmark-level contamination [oren2024proving]. We include ablations that preempt the standard confounds: deduplicated versus non-deduplicated Pythia (duplication), frequency-matched member/non-member splits (string frequency), and model-size scaling (does the contamination→leakage link strengthen with scale, as memorization does [carlini2023quantifying]). Differentially private training [abadi2016deep,li2022dpllm] is discussed as the mitigation direction (Section

This section fixes the threat model, methods, data, and metrics; the empirical results under this protocol—per-detector TPR at low FPR with log-scale ROC, extraction rates, and the headline contamination→leakage correlation with confidence intervals—are reported in the results section, with every reported number tracing to a logged harness run.

Results

All results in this section are preliminary, obtained on Pythia-160M on CPU with $N = 300$ ground-truth Pile members (seed 0); larger-model rows are left for the GPU replication. Every number is reproducible from a seeded script and recorded in our results ledger.

Membership separation is at chance on a confound-clean split

We first reproduce, as a control, the known weakness of membership inference on pre-trained LLMs [duan2024mia]. On a confound-clean split (members = Pile train, non-members = Pile validation, stratified across 22 Pile subsets to match domain), all four detectors sit at chance at

160M (Table [1](#tab:membership)); on the temporally-confounded WikiMIA split the same model shows a spurious 0.52–0.56, and a 1.4B model rises further—evidence that the WikiMIA signal is substantially distribution shift, not membership.

<div id="tab:membership">

Construction (model)	LOSS	Min-K%	Min-K%++	zlib
Pile train-vs-val, clean (160M)	0.454	0.470	0.490	0.484
WikiMIA-64, confounded (160M)	0.523	0.539	0.545	0.564
WikiMIA-64, confounded (1.4B)	0.571	0.580	0.547	0.616

Membership AUC. Chance (≈ 0.5) on the confound-clean split at 160M; the WikiMIA “signal” is largely temporal/topical distribution shift. CIs in the ledger; deduplicated Pythia gives the same chance-level result.

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Contamination predicts leakage — but only through loss

Our headline analysis correlates each per-item detector score with the per-item extraction outcome (prefix-continuation extractable memorization under greedy decoding [carlini2023quantifying]), then controls for raw loss. Table [2](#tab:headline) reports, for each calibrated detector, the zero-order Spearman ρ , the linear partial ρ given loss, the non-linear (cubic-residual) partial ρ with bootstrap CI, the FDR-corrected permutation q , and the mediation decomposition.

<div id="tab:headline">

Detector	zero-order	partial loss	cubic-resid. [95 % CI]	BH-q	mediation: direct indirect
LOSS	+ 0.275	—	—	—	(mediator)
Min-K%	+ 0.173	− 0.178	− 0.110 [− 0.234, − 0.002]	0.058	− 0.394 + 0.567
Min-K%++	+ 0.108	− 0.148	− 0.160 [− 0.287, − 0.041]	0.015	− 0.213 + 0.321
zlib	+ 0.177	− 0.042	− 0.052 [− 0.165, + 0.068]	0.331	− 0.061 + 0.238

Headline: per-item contamination score vs. extraction (Spearman ρ), Pythia-160M, $N = 300$ members. The positive zero-order correlations collapse to ≈ 0 or significantly *negative* once loss is controlled—linearly, and under the non-linear cubic-residual control (no positive signal revives; deciles and the deduplicated arm agree). Mediation: the loss-mediated *indirect* effect is significantly positive for all three detectors while the *direct* effect is null (zlib) or negative (Min-K%, Min-K%++). We read this as a *descriptive* decomposition, not a causal mediation claim (see below): no calibrated detector adds positive signal beyond loss.

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Collinearity caveat (why we do not over-read the negative partials).

The calibrated detectors are deterministic transforms of the same per-token log-probabilities as loss, and are empirically collinear with it: Spearman $\rho(\text{loss}, \cdot) = 0.90$ (Min-K%), 0.74 (Min-K%++), 0.74 (zlib), with variance-inflation factors 6.2, 2.6, 2.4. The strongest negative partial (Min-K%, the most loss-collinear detector at VIF 6.2) is therefore consistent with a *suppression artifact* of near-collinearity rather than substantive inverse prediction; we do not claim the calibrated detectors *negatively* predict leakage. The defensible, conservative statement is that they carry *no positive* leakage signal independent of loss. Min-K%++ and zlib have only moderate collinearity (VIF ≤ 3), so their null/near-null residuals are less attributable to collinearity.

The pre-registered decision rule asked whether any calibrated detector predicts leakage *beyond* loss (a positive partial ρ , CI excluding zero, FDR-significant). None does, under the linear or the non-linear control. **Power note:** with $N = 300$ and a near-degenerate outcome (3/300 fully extracted), this is evidence of *no positive independent signal of appreciable size*, not proof of an exact null; the analysis is well-powered only for moderate-to-large positive residuals, and a small positive effect at scale is not excluded (hence the GPU replication). The per-domain breakdown (ledger) shows the loss $\leftrightarrow$ extraction link is heterogeneous and sign-flipping across domains—strongest in templated/structured domains (GitHub, StackExchange), reversed in some prose domains (PubMed Abstracts)—so the pooled ρ is a domain-mixture, not a uniform effect.

Extraction and PII at this scale

Extractable memorization is rare at 160M: 3/300 members are fully extractable (exact-match extraction rate 0.010; mean fractional extraction 0.037), the fully-extracted items being templated boilerplate. On the Enron-Emails-in-Pile subset we measured *zero* verbatim PII leakage (8/36 documents contained PII in the held suffix; none were regurgitated). We report the PII result as a null at this scale and make no PII-exposure claim; both quantities are expected to grow with model scale.

Benchmark contamination (model-free *n*-gram + permutation test)

We complement the per-item analysis with two benchmark-level contamination tests (Table [data-reference-type="ref" data-bbox="674 603 889 616" style="font-family: monospace;">data-reference="tab:matrix">3](#tab:matrix)). The model-free *n*-gram overlap against a public *sample* of the Pile (10k documents) is a scale-invariant method but, with a sampled reference, yields only a loose *lower bound*: overlap is near-zero for MMLU (0.2% at 13-grams), GSM8K (0%), and HumanEval (0% at 13-grams), which certifies overlap is *at least* this small and is uninformative about true contamination—a full-Pile index (infrastructure-, not GPU-, gated) is required for a real rate. The Oren permutation/exchangeability test [oren2024proving] at 160M finds the canonical ordering favoured beyond chance for MMLU ($p = 0.001$) and GSM8K ($p = 0.013$) but not HumanEval ($p = 0.875$); we draw *no* contamination conclusion from this, as the test is membership-based, run at sanity scale (small k , smallest model), and subject to a fluency/orientation artifact—it is flagged GPU-gated and requires a fluency-control baseline before any claim.

<div id="tab:matrix">

Benchmark	13-gram overlap (lower bound)	8-gram overlap	Oren p (160M, sanity)
MMLU	0.2%	0.8%	0.001
GSM8K	0.0%	0.0%	0.013

HumanEval	0.0%	1.8%	0.875
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Benchmark-level contamination at small scale. n -gram cells are a *lower bound* against a 10k Pile sample (method scale-invariant, reference under-powered); Oren p -values are sanity-scale at 160M and GPU-gated (no contamination conclusion drawn). See docs/contamination_matrix.md.

</div>

Discussion

Membership detection and leakage prediction diverge.

The central empirical observation is that the contamination/membership signal which predicts *extraction* is, to the resolution of our experiment, *just raw loss*. The reference-free detectors that the contamination-detection literature has invested in—Min-K%, Min-K%++, zlib—improve membership ranking by re-calibrating the per-token likelihood (z-scoring against the vocabulary, compressing, or trimming to the lowest-probability tokens), but in doing so they discard precisely the loss-magnitude information that tracks how extractable an item is. A descriptive mediation decomposition is consistent with this—the loss-mediated (indirect) path is positive for all three detectors while the direct paths are null or negative—but we read it descriptively, not causally: the detectors are near-collinear transforms of loss (Spearman up to 0.90; VIF up to 6.2), so a negative direct/partial term is consistent with statistical suppression rather than genuine inverse prediction. We therefore claim only the conservative version: the calibrated detectors add *no positive* leakage signal beyond loss. A practitioner who wants to know *which contaminated items the model will actually leak* is, on this evidence, no better served by a state-of-the-art membership detector than by raw loss. This is the sense in which membership detection and leakage prediction are different tasks.

Why this is a security result, not a leaderboard result.

Our finding is deliberately *not* “we built a better detector.” It is that the privacy question—will contamination of a benchmark expose a leakage channel?—is mis-served by importing the membership-inference toolkit wholesale. For an auditor of a released model, the actionable implication is to measure loss/extractability directly and to treat a high Min-K%/Min-K%++ score as evidence about membership, not about leakage risk. This reframing is the contribution; the detectors themselves are prior work.

Relation to concurrent work.

Our direction agrees with two recent results and we do not claim the bottom line is surprising: Al Sahili et al. [alsahili2025effectiveness] report only “marginal” gains of MIA scores over likelihood ranking for targeted extraction, and Hayes et al. [hayes2025strong] find no correlation between (LiRA) membership success and extraction. We add the controlled, mechanistic form of the claim—a pre-registered partial-correlation/mediation that quantifies a *zero-to-negative* residual for the calibrated reference-free detectors after loss is removed—and we target the reference-free detectors the contamination literature actually deploys rather than a shadow-model attack. Chen et al. [chen2025statistical] independently find these detectors do not robustly beat the loss baseline for *membership* once seed variance is accounted for; our result is the extraction-outcome analogue.

Defenses.

Because the leakage we measure is downstream of memorization, the principled mitigation is differential privacy applied at training time [abadi2016deep,li2022dpllm]; it is a producer-side

control, not an auditor-side detector, and bounds the very quantity (loss-magnitude / memorization) our analysis identifies as the operative one.

Limitations

We state the limitations plainly; several bound the strength of the present claims and motivate the GPU-scale replication the pipeline is built for.

- **Single, smallest model.** All results are on Pythia-160M (CPU). Memorization grows log-linearly with model scale [carlini2023quantifying], so both the membership signal and the extraction outcome are expected to be stronger at 1.4B–12B. The present numbers are *preliminary*; we have built every analysis so the larger-model run is a one-line configuration change.
- **Chance-level membership separation.** On the confound-clean Pile train-vs-val split, membership AUC is at chance (0.45–0.49) at 160M, consistent with [duan2024mia]. The divergence result is therefore established in a regime where the membership signal is itself weak; whether the calibrated detectors gain *independent* leakage-predictive value once membership separation becomes non-trivial at scale is an open question our design is poised to answer.
- **Near-degenerate extraction outcome.** Extractable memorization at 160M is rare (3/300 items fully extracted; mean fractional extraction 0.037), so the correlation analysis leans on a small high-extraction tail. We mitigate with rank statistics, bootstrap CIs, and a zero-robust Kendall check, but a less zero-inflated outcome at scale would sharpen all estimates.
- **PII not yet demonstrated.** On the Enron-in-Pile subset we observed *zero* verbatim PII leakage at 160M (8/36 documents contained PII in the held suffix; none were regurgitated). The PII limb of the threat model is thus a designed capability with a null result at this scale, not a demonstrated leak; we report it as such and do not claim PII exposure.
- **Benchmark-level test underpowered.** The Oren permutation/exchangeability test is run only at sanity scale on 160M; membership-based, it is underpowered here and is flagged as GPU-gated rather than used to draw contamination conclusions.
- ***n*-gram contamination is a lower bound.** Our model-free *n*-gram overlap uses a public *sample* of the Pile as the reference index, so measured benchmark↔Pile overlap underestimates the true overlap against the full corpus.
- **Observational, members-only correlation.** The headline analysis correlates detector scores with extraction across known members; it is observational, not interventional. We address the most important confound (loss) by pre-registered partial correlation and mediation, and the obvious alternatives (frequency, duplication, non-linearity, distribution shift) by explicit controls, but residual confounding cannot be excluded.
- **Collinearity of detectors with loss.** The calibrated detectors are deterministic transforms of the same per-token log-probabilities as loss and are empirically collinear with it (Spearman 0.74–0.90; VIF up to 6.2 for Min-K%). Consequently we interpret the negative partial/direct terms as possible *suppression artifacts* of near-collinearity and claim only the conservative “no positive residual” result; we do not assert the detectors inversely predict leakage.

- **Construct validity of the leakage proxy.** The outcome (greedy prefix-continuation extraction over the held suffix) is itself likelihood-related, so part of the loss↔ extraction association is mechanical/definitional. Our control removes the loss component, but a decisive separation would compute prefix-only loss against extraction; we flag this as a known construct-validity limitation rather than claiming the two are independent by construction.
- **Selection and aggregation.** Members are drawn from a non-uniform public Pile sample (pile-10k), so member-selection bias is possible; and the pooled correlation aggregates domains whose effects flip sign (Section [6.2](#sec:res-headline)), so the pooled ρ should be read as a domain-mixture, not a homogeneous effect.
- **Linearity (now addressed).** An earlier version controlled for loss only linearly; we added a cubic-residual and decile-stratified non-linear control, under which no positive independent signal revives. We note it here because it was a live threat to the claim until tested.

Conclusion

We argued that benchmark contamination is best understood as a privacy/security vulnerability and asked, on models with ground-truth public training data, whether the contamination/membership signal that a benchmark leaks actually predicts concrete extraction. Using a pre-registered partial-correlation and mediation analysis that controls for raw per-item loss, we found that it does—but only through loss: the calibrated reference-free detectors (Min-K%, Min-K%++, zlib) add no independent predictive value beyond loss, and two are negatively associated with extraction once loss is held fixed. The result is robust to a non-linear loss control and to deduplication, and is not a frequency or zero-inflation artifact. The practical message is a divergence: the detectors optimized for membership inference are not the right instrument for the leakage question, and an auditor should measure loss/extractability directly. We claim no new detector or metric; the contribution is the security reframing and the controlled, pre-registered measurement. These findings are preliminary, on the smallest Pythia model; the immediate next step—and the design target of our released pipeline—is the GPU-scale replication across model sizes, where memorization, extraction, and any PII leakage are expected to strengthen, and where the question of whether calibrated detectors gain independent leakage-predictive value at scale can be settled.